BACHELOR OF COMPUTER APPLICATIONS (BCA)

SEMESTER VI - END SEMESTER EXAMINATION

May 2024

Paper Code: BCA-609

Subject: Python Programming and Data Science

Time Allowed: 3 Hours | Maximum Marks: 75

General Instructions:

1. Attempt all sections. Section A is compulsory; internal choice is given in Sections B and C.
2. Write answers in legible handwriting and number them exactly as in the question paper.
3. Draw neat, labelled diagrams wherever required.

SECTION A (5 x 3 = 15 Marks)

Answer all five questions. Each question carries 3 marks.

1. Differentiate between a list and a tuple in Python. [3]

2. What is a dictionary in Python? [3]

3. Explain the purpose of pandas. [3]

4. What is a NumPy array? [3]

5. Define a lambda function. [3]

SECTION B (4 x 10 = 40 Marks)

Answer any four questions. Each question carries 10 marks.

1. Explain Python data types and type conversion with examples. [10]

2. Explain file handling in Python with examples. [10]

3. Explain exception handling using try, except and finally. [10]

4. Explain data cleaning using pandas. [10]

SECTION C (2 x 10 = 20 Marks)

Answer any two questions. Each question carries 10 marks.

1. Write and explain a Python program that reads a CSV file and plots a bar chart. [10]

2. Explain modules, packages and virtual environments in Python. [10]

*** *END OF PAPER* ***